

PEE question set 1

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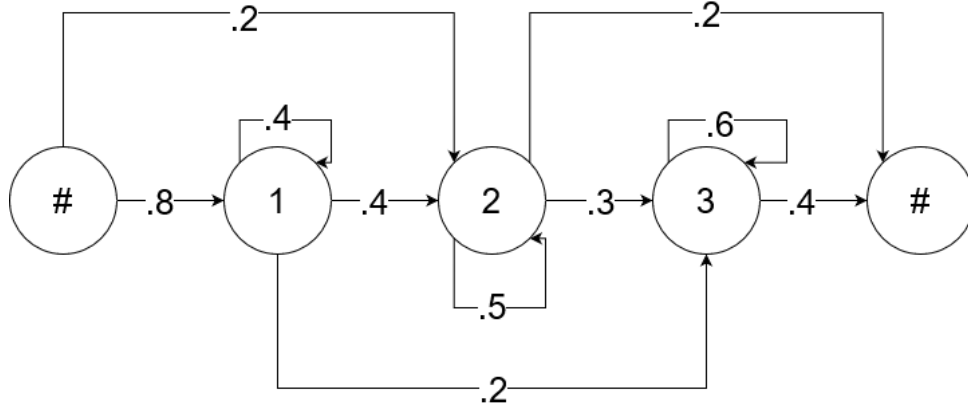
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1 Question 1

The probability of having cavities if there are holds between the teeth and there is no pain can be represented by the following equation:

$$\begin{aligned}
 P(C = si | H = si, D = no) &= \frac{P(C = si, H = si, D = no)}{P(H = si, D = no)} = \\
 \frac{P(C = si, H = si, D = no)}{\sum_{x \in \{si, no\}} P(C = x, H = si, D = no)} &= \frac{P(C = si)P(H = si | C = si)P(D = no | C = si)}{\sum_{x \in \{si, no\}} P(C = x)P(H = si | C = x)P(D = no | C = x)} = \\
 \frac{0.3 \times 0.9 \times 0.3}{(0.3 \times 0.9 \times 0.3) + (0.7 \times 0.2 \times 0.9)} &= \frac{0.081}{0.081 + 0.126} = 0.391
 \end{aligned}$$

2 Question 2



$$Q = \{1, 2, 3\}; \Sigma = \{a, b\}$$

u = 1	o = a	o = b
u = 2	0.2	0.8
u = 2	0.8	0.2
u = 2	0.5	0.5

We assume that $P(r_j | r_1, r_2, \dots, r_{j-1}) = P(r_j) = q(u_j | v_{j-1}) \times e(o | u_j)$:

- (1) $S \rightarrow aX = q(1|\#) \times e(a|1) = 0.8 \times 0.2 = 0.16$
- (2) $S \rightarrow bX = q(1|\#) \times e(b|1) = 0.8 \times 0.8 = 0.64$
- (3) $S \rightarrow aY = q(2|\#) \times e(a|2) = 0.2 \times 0.8 = 0.16$
- (4) $S \rightarrow bY = q(2|\#) \times e(b|2) = 0.2 \times 0.2 = 0.04$
- (5) $X \rightarrow aX = q(1|1) \times e(a|1) = 0.4 \times 0.2 = 0.08$
- (6) $X \rightarrow bX = q(1|1) \times e(b|1) = 0.4 \times 0.8 = 0.032$
- (7) $X \rightarrow aY = q(2|1) \times e(a|2) = 0.4 \times 0.8 = 0.032$
- (8) $X \rightarrow bY = q(2|1) \times e(b|2) = 0.4 \times 0.2 = 0.08$
- (9) $X \rightarrow aZ = q(3|1) \times e(a|3) = 0.2 \times 0.5 = 0.1$
- (10) $X \rightarrow bZ = q(3|1) \times e(b|3) = 0.2 \times 0.5 = 0.1$
- (11) $Y \rightarrow aY = q(2|2) \times e(a|2) = 0.5 \times 0.8 = 0.4$
- (12) $Y \rightarrow bY = q(2|2) \times e(b|2) = 0.5 \times 0.2 = 0.1$
- (13) $Y \rightarrow aZ = q(3|2) \times e(a|3) = 0.3 \times 0.5 = 0.15$
- (14) $Y \rightarrow bZ = q(3|2) \times e(b|3) = 0.3 \times 0.5 = 0.15$
- (15) $Y \rightarrow \epsilon = q(\#|2) = 0.2$
- (16) $Z \rightarrow aZ = q(3|3) \times e(a|3) = 0.6 \times 0.5 = 0.3$
- (17) $Z \rightarrow bZ = q(3|3) \times e(b|3) = 0.6 \times 0.5 = 0.3$
- (18) $Z \rightarrow \epsilon = q(\#|3) = 0.4$

To generate “aba” with the given PCFG:

t_1 : (1) aX → (8)abY → (13)abaZ → (18)aba = $P(\text{aba}|t_1) = 0.16 \times 0.08 \times 0.15 \times 0.4 = 7,68 \times 10^{-4}$

t_2 : (3) aY → (14)abZ → (16)abaZ → (18)aba = $P(\text{aba} | t_2) = 0.16 \times 0.15 \times 0.3 \times 0.4 = 2.9 \times 10^{-3}$

t_3 : (3) aY → (12)abY → (11)abaY → (15)aba = $P(\text{aba} | t_3) = 0.16 \times 0.1 \times 0.4 \times 0.2 = 1.3 \times 10^{-3}$

3 Question 3

For this question we have to come up with an algorithm that maximizes the probability of the parsing tree having in mind a past history with size two (trigrams).

The definition for this trigram generalization will be:

$$\gamma_t(s, s') = \max_{y_1^t: y_t = s} \prod_{i=2}^t \Psi_i(y_{i-2}, y_{i-1}, y_i, x_i) = \max_{y_1^t: y_t = s} \prod_{i=2}^t q(y_i | y_{i-2}, y_{i-1}) e(x_i | y_i)$$

Note that we have added a new parameter to the gamma function. This will represent the previous state and will be used to compute the past history. Knowing this, the initialization step must change. In this case we have to initialize both instant $t = 1$ and $t = 2$ since both lack a past history with size 2.

$$\gamma_1(s, \#) = \Psi_1(y_{-1} = \#, y_0 = \#, y_1 = s, x_1)$$

$$\gamma_2(s, s') = \gamma_1(s', \#) \Psi_2(y_0 = \#, y_1 = s', y_2 = s, x_2)$$

With this, we have computed the first initial steps. So, the recursion step can be defined as follows: $\forall s, s' \in Y$

$$\gamma_t(s, s') = \max_{s'' \in Y} \{ \gamma_{t-1}(s', s'') \Psi_t(y_{t-2} = s'', y_{t-1} = s', y_t = s, x_t) \}$$

Thus, the maximum probability for a sequence with this trigram model will be:

$$\max_{s, s' \in Y} \gamma_t(s, s')$$

4 Question 4

In this skip-gram problem, we are omitting the previous state when computing the actual. Instead, we are using the two times previous state: $t-2$ for instant t . We're doing this for the transition scores, since the emission only depends on the actual state. Having this in mind, we define this skip-gram model as follows:

$$\alpha_t(s, s') = \sum_{y_1^t: y_t = s} \prod_{i=1}^T \Psi_i(y_{i-2}, y_i, x_i)$$

As expected, the initialization changes, having to define two initial states for time 1 and 2: $\forall s \in Y$

$$\alpha_1(s, \#) = \Psi_1(y_{-1} = \#, y_1 = s, x_1)$$

$$\alpha_2(s, s') = \Psi_2(y_0 = \#, y_2 = s, x_2) \alpha_1(s', \#)$$

As we can see, we have defined α_1 in terms of the actual state and null, since there's no previous state nor two times previous. For α_2 things change. There we have a previous state but not a two times previous state, thus we pass to the Psi function null, and then compute the observation for the existing previous state.

Generalizing the previous case for the rest, we came up with the following definition: $\forall t : 3 \dots T; s \in Y$

$$\alpha_t(s, s') = \sum_{s'' \in Y} \alpha_{t-1}(s', s'') \Psi_t(y_{t-2} = s'', y_t = s, x_t)$$

Finally, the probability of the observation x_1^T is:

$$\sum_{s, s' \in Y} \alpha_T(s, s')$$

5 Question 5

To compute the number of parsing trees that a grammar produces for a certain sequence, we define a function π as follows:

The initial state will be computed by checking all the non-terminal to terminal productions and setting them to 1 if the non-terminal produces the required terminal

$$\pi(A, i, i + 1) = \mathbb{1}(A \rightarrow b)\delta(b, x_{i+1})$$

For the recursion step, we understand that, for a given rule, for example A producing BC, being all of them non-terminals, the number of derivation trees can be computed as the sum of the number of trees that B produces times the number of trees that C produces for each one of the possible length of the sequence. Just as the following expression says.

$$\pi(A, i, i + l) = \sum_{B, C \in \{N\}} \mathbb{1}(A \rightarrow BC) \left(\sum_{k=1}^{l-1} \pi(B, i, i + k) \times \pi(C, i + k, i + l) \right)$$

This algorithm returns a scalar representing the number of derivation trees that can be produced for a certain sequence, and tested with the given grammar and the one seen in class we see that it returns 2. The correct answer.

6 Question 6

For a left branching grammar, we see that its main difference with normal grammar is that at each step, this left-branching must produce at the most right position a terminal. This implication does not modify the initialization state of the CKY-based *Viterbi* algorithm. Which remains:

$$\hat{e}(A < i, i + 1 >) = P(A \rightarrow b)\delta(b, x_{i+1})$$

The recursion step can be written as follows: $\forall A \in N; \forall l : 2..T; \forall i : 0..T - l$ being T the length of the sequence

$$\hat{e}(A < i, i + l >) = \max_{B, C \in N} P(A \rightarrow BC) \hat{e}(B < i, i + l - 1 >) \hat{e}(C < i + l - 1, i + l >)$$

Finally, the best decision tree can be represented by:

$$\hat{P}_{G_\theta}(x) = \hat{e}(S < 0, n >)$$

Note that the principal difference between this algorithm and a regular CKY-based *Viterbi* algorithm is that the value of the k is always $k = l - 1$. Thus, the time complexity is reduced to $\Theta(n^2)$